Michael Ichikawa

AI ENGINEER · AGENTIC SYSTEMS

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SUMMARY

I shipped a production AI agent solo, from an empty repo to a listing on the Microsoft Commercial Marketplace and the Teams Store, doing the product, the engineering and the compliance myself. I built a multi-agent workflow engine from scratch for the same class of problem a level up, composing single-purpose agents into a validated graph, with a human approval gate before any step that acts. Before all of it, MS Mathematics, eight years teaching undergraduate math, and three years of semiconductor physical design at Intel.

FLAGSHIP WORK

Microclaw — Production AI Agent for Microsoft 365

Microsoft Marketplace listing

One agent across Microsoft 365, with 87 function-calling tools and a tool-calling loop that chains them until a task is complete. The hard part was choosing: sending every tool on every message is expensive, so a retrieval step narrows the list first, and I measure how often that drops a tool the request needed. Around that: RAG semantic memory, model routing that escalates to a larger model only when a turn needs it, and multi-layer failure recovery.

TypeScript · Azure OpenAI · Microsoft Graph · Azure Postgres (RLS) · Bicep

Multi-Agent Workflow Engine — built from scratch

mtichikawa.github.io/workflow-engine

A Python engine, built from scratch on the standard library, that turns a plain-English request into a graph of single-purpose agents. Every workflow is statically validated before it runs, and any step that acts outside the engine must sit behind a human approval gate. A learning loop infers a team's unwritten rules from their corrections, with no fine-tuning: on 150 blind-written messages, routing accuracy rose from 59 to 97 percent. On 320 blind-written patient messages, its triage step sent all 160 medical ones to a person, with no false alarms.

Python · agentic orchestration · MCP (client + server) · contracts + recipes · static graph validation

TECHNICAL SKILLS

AI-Native Dev	directing AI agents to implement, test and debug complex features · Claude Code (daily driver) · agent-directed builds verified by tests and evals · per-repo context files · AI-native development practices
Agents / LLM	agent orchestration · tool-calling agent loops · function calling · tool use · tool retrieval (KNN) · RAG (retrieval-augmented generation) · vector search & embeddings (pgvector · FAISS · BigQuery Vector Search) · semantic memory · multi-agent workflows · human-in-the-loop gating · MCP (client + server) · in-context learning
Reliability & Cost	failure-mode engineering (backoff/retry · context compaction · loop detection · abort ceilings) · model routing with mid-task escalation · input-token reduction · multi-tenant isolation (Postgres RLS)
Evaluation	eval design · expected-facts scoring · A/B against baselines · blind held-out test sets · pytest · Vitest · GitHub Actions CI
Built with	Python · TypeScript · Node.js · SQL · Azure OpenAI · Anthropic API · Microsoft Graph · PostgreSQL · BigQuery · Delta Lake (medallion) · Spark Structured Streaming · SQLAlchemy · pandas · scikit-learn · HuggingFace · sentence-transformers · Azure (Container Apps · Entra · Bicep) · AWS (S3 · Lambda · DynamoDB) · GCP (Vertex AI · Cloud Run) · Docker · FastAPI / REST · Redis

EXPERIENCE

Founding AI/ML Engineer

2026 – present

Microclaw LLC · Portland, OR

- Built the agent loop and redesigned its tool selection: on 362 blind-written test requests it offered every tool a request needed 98.1 percent of the time, up from 87.0, and on a 142-case agent eval it called the right tools on 134 of 142 cases on average over three runs, with 49 percent fewer prompt tokens than sending every tool
- About ten weeks from first commit to a Marketplace listing, solo, through Microsoft Marketplace certification and the Fulfillment and Metering API work that makes it billable, then Teams Store validation after seven review rounds
- Multi-tenant isolation via Postgres row-level security: FORCE policies on every tenant-scoped table, an app role that cannot bypass them, and delegated Entra auth

Career Break — Full-Time Parent · Self-Directed Transition to Software

2023 – 2026

Portland, OR

- Three years as primary caregiver, retraining into software throughout
- Started in data engineering and ML, then moved to agentic development as the field did

Mathematics Faculty

2015 – 2023

Santa Rosa Junior College · Santa Rosa, CA

- Taught mathematics for eight years at an open-enrollment college: Algebra through Calculus III, Linear Algebra, Differential Equations and introductory Statistics
- Wrote my own lectures and exams, found the topics students kept missing, and reworked how I taught them
- Wrote my own version of an open-source precalculus textbook in LaTeX, reorganized by concept so trig functions came in week 2 instead of week 10

Physical Design Engineer

2008 – 2011

Intel Corporation · Semiconductor R&D

- Laid out blocks of the Ivy Bridge (22nm) microprocessor, closing timing, power and design-rule checks
- Kept my blocks passing every check as cell-library updates and upstream circuit changes landed in a large, version-controlled shared design
- Wrote Python and Perl automation for physical verification

EDUCATION

MS Mathematics — The City College of New York — 2014

BS Mechanical Engineering — University of California, Berkeley — 2008

PORTFOLIO PROJECTS — full portfolio at mtichikawa.github.io

Self-directed work from the 2023–2026 transition above, before the move to agentic systems. The one closest to current work is a **RAG pipeline on GCP** with three interchangeable retrievers behind one interface, BigQuery Vector Search, FAISS and pgvector, so they could be compared on the same eval.

The rest spans data engineering and ML: streaming and lakehouse pipelines · cloud ETL · anomaly detection · LLM tooling and bias testing · forecasting · a trading backtest system end to end.